INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II ,First Semester Examination, October- 2025

Subject Code: BCACC103

Subject Name: Discrete Structure

Full Marks: 30 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

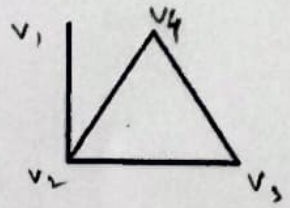
1. Draw a tree with five vertices and total degree 8. 2
2. Define Binary Tree with proper example. 2
3. Define Spanning Tree.
4. Find the number of pendant vertices in a binary tree with 11 vertices. 2
5. How many edges a complete graph has with 10 vertices? 2
6. The sum of the degrees of all the vertices of a graph is 40. Find the number of edges of the graph. 2

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Draw the graph whose incidence matrix is $\begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$ 5

8. Construct the graph whose adjacency matrix is $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ 5

9. Determine the adjacency matrix of the following graph : 5

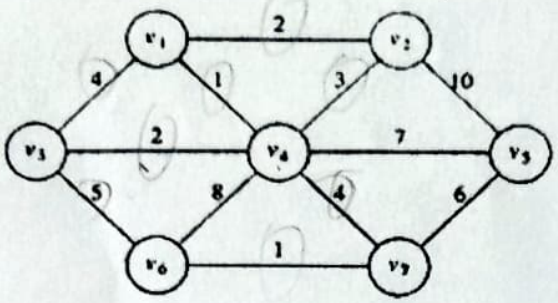


Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Using Kruskal's Algorithm find the minimal spanning tree and it's total weight from the following graph **10**



11. Using Prim's Algorithm find the minimal spanning tree and it's total weight from the following graph **10**

